

Publication List

Paul M. Chichura

Last updated: May 2026

Metrics. I have authored 33 publications, including two first-author publications. These publications have received 951 citations (H-Index: 16), according to [NASA ADS](#).

Journal abbreviations. A&A: [Astronomy & Astrophysics](#); ApJ: [The Astrophysical Journal](#); ApJL: [The Astrophysical Journal Letters](#); JCAP: [Journal of Cosmology and Astroparticle Physics](#); MNRAS: [Monthly Notices of the Royal Astronomical Society](#); PhRvD: [Physical Review D](#); PhRvL: [Physical Review Letters](#).

Note on Authorship. I am a member of the [South Pole Telescope \(SPT-3G\)](#) and [Event Horizon Telescope \(EHT\)](#) collaborations, which follow authorship policies wherein all contributing members are credited. I list all publications on which I am a co-author, and I separate my first-author publications to highlight my primary individual contributions.

(1) First-Author Publications

- [1] **Chichura, P. M.**, A. Rahlin, A. J. Anderson, et al. 2025. “Pointing Accuracy Improvements for the South Pole Telescope with Machine Learning.” *Journal of Astronomical Instrumentation* 14 (January): 2550001. DOI: [10.1142/S2251171725500011](#). [arXiv:2412.15167](#).
- [2] **Chichura, P. M.**, A. Foster, C. Patel, et al. 2022. “Asteroid Measurements at Millimeter Wavelengths with the South Pole Telescope.” *ApJ* 936, no. 2 (September): 173. DOI: [10.3847/1538-4357/ac89ec](#). [arXiv:2202.01406](#).

(2) Significant Contributions

- [3] Wan, Y., J. D. Vieira, **Chichura, P. M.**, et al. 2026. “Detection of Millimeter-wavelength Flares from Two Accreting White Dwarf Systems in the SPT-3G Galactic Plane Survey.” *ApJ* 997, no. 2 (February): 133. DOI: [10.3847/1538-4357/ae2de8](#). [arXiv:2509.08962](#).
- [4] Kohn, S. A., J. E. Aguirre, P. La Plante, et al. 2019. “The HERA-19 Commissioning Array: Direction-dependent Effects.” *ApJ* 882, no. 1 (September): 58. DOI: [10.3847/1538-4357/ab2f72](#). [arXiv:1802.04151](#).

(3) All Publications

- [5] Archipley, M., A. Hryciuk, L. E. Bleem, et al. 2026. “Millimeter-wave observations of Euclid Deep Field South using the South Pole Telescope: A data release of temperature maps and catalogs.” *A&A* 706 (January): A17. DOI: [10.1051/0004-6361/202555798](#). [arXiv:2506.00298](#).
- [6] Bernshteyn, V., N. S. Conroy, M. Bauböck, et al. 2026. “Ring Asymmetry and Spin in M87*.” *ApJ* 1000, no. 2 (April): 231. DOI: [10.3847/1538-4357/ae34af](#). [arXiv:2601.00394](#).
- [7] Camphuis, E., W. Quan, L. Balkenhol, et al. 2026. “SPT-3G D1: CMB temperature and polarization power spectra and cosmology from 2019 and 2020 observations of the SPT-3G main field.” *PhRvD* 113, no. 8 (April): 083504. DOI: [10.1103/7wt3-9v2y](#). [arXiv:2506.20707](#).
- [8] Chaubal, P., N. Huang, C. L. Reichardt, et al. 2026. “SPT-3G D1: A Measurement of Secondary Cosmic Microwave Background Anisotropy Power.” *arXiv e-prints* (January): arXiv:2601.20551. DOI: [10.48550/arXiv.2601.20551](#). [arXiv:2601.20551](#).

- [9] de Haan, T., M. Archipley, N. Huang, et al. 2026. “Characterization of the Polarization Beam Response of SPT-3G Using Point Sources.” *arXiv e-prints* (February): arXiv:2602.06334. DOI: [10.48550 / arXiv.2602.06334](https://doi.org/10.48550/arXiv.2602.06334). [arXiv:2602.06334](https://arxiv.org/abs/2602.06334).
- [10] Georgiev, B., P. Tiede, S. D. von Fellenberg, et al. 2026. “Locating the missing large-scale emission in the jet of M87* with short EHT baselines.” *A&A* 709 (April): A26. DOI: [10.1051/0004-6361/202557013](https://doi.org/10.1051/0004-6361/202557013). [arXiv:2601.13356](https://arxiv.org/abs/2601.13356).
- [11] Maniyar, A. S., F. Bianchini, W. L. K. Wu, et al. 2026. “SPT-3G D1: Compton- y maps using data from the SPT-3G and Planck surveys.” *arXiv e-prints* (February): arXiv:2602.11279. DOI: [10.48550 / arXiv.2602.11279](https://doi.org/10.48550/arXiv.2602.11279). [arXiv:2602.11279](https://arxiv.org/abs/2602.11279).
- [12] Qu, F. J., F. Ge, W. L. K. Wu, et al. 2026. “Unified and Consistent Structure Growth Measurements from Joint ACT, SPT, and Planck CMB Lensing.” *PhRvL* 136, no. 2 (January): 021001. DOI: [10.1103 / k5yr-3h6d](https://doi.org/10.1103/k5yr-3h6d). [arXiv:2504.20038](https://arxiv.org/abs/2504.20038).
- [13] Quan, W., E. Camphuis, C. Daley, et al. 2026. “SPT-3G D1: Maps of the millimeter-wave sky from 2019 and 2020 observations of the SPT-3G Main field.” *arXiv e-prints* (March): arXiv:2603.20163. DOI: [10.48550/arXiv.2603.20163](https://doi.org/10.48550/arXiv.2603.20163). [arXiv:2603.20163](https://arxiv.org/abs/2603.20163).
- [14] Raghunathan, S., P. A. R. Ade, D. Anbajagane, et al. 2026. “Measurement of the Full Shape of the Thermal SunyaevZel’dovich Power Spectrum from the South Pole Telescope and HerschelSPIRE Observations.” *ApJL* 1002, no. 1 (May): L22. DOI: [10.3847/2041-8213/ae5c07](https://doi.org/10.3847/2041-8213/ae5c07). [arXiv:2602.10107](https://arxiv.org/abs/2602.10107).
- [15] Saurabh, H. Müller, S. D. von Fellenberg, et al. 2026. “Probing jet base emission of M87* with the 2021 Event Horizon Telescope observations.” *A&A* 706 (January): A27. DOI: [10.1051/0004-6361/202557022](https://doi.org/10.1051/0004-6361/202557022). [arXiv:2512.08970](https://arxiv.org/abs/2512.08970).
- [16] Tandoi, C., A. Foster, T. J. Maccarone, et al. 2026. “Millimeter-wave Detections of Symbiotic Stars in SPT and ACT Data.” *arXiv e-prints* (May): arXiv:2605.01022. DOI: [10.48550 / arXiv.2605.01022](https://doi.org/10.48550/arXiv.2605.01022). [arXiv:2605.01022](https://arxiv.org/abs/2605.01022).
- [3] Wan, Y., J. D. Vieira, **Chichura, P. M.**, et al. 2026. “Detection of Millimeter-wavelength Flares from Two Accreting White Dwarf Systems in the SPT-3G Galactic Plane Survey.” *ApJ* 997, no. 2 (February): 133. DOI: [10.3847/1538-4357/ae2de8](https://doi.org/10.3847/1538-4357/ae2de8). [arXiv:2509.08962](https://arxiv.org/abs/2509.08962).
- [1] **Chichura, P. M.**, A. Rahlin, A. J. Anderson, et al. 2025. “Pointing Accuracy Improvements for the South Pole Telescope with Machine Learning.” *Journal of Astronomical Instrumentation* 14 (January): 2550001. DOI: [10.1142/S2251171725500011](https://doi.org/10.1142/S2251171725500011). [arXiv:2412.15167](https://arxiv.org/abs/2412.15167).
- [17] Coerver, A., J. A. Zebrowski, S. Takakura, et al. 2025. “Measurement and Modeling of Polarized Atmosphere at the South Pole with SPT-3G.” *ApJ* 982, no. 1 (March): 15. DOI: [10.3847/1538-4357/ada35d](https://doi.org/10.3847/1538-4357/ada35d). [arXiv:2407.20579](https://arxiv.org/abs/2407.20579).
- [18] Event Horizon Telescope Collaboration et al. 2025. “Horizon-scale variability of M87* from 20172021 EHT observations.” *A&A* 704 (December): A91. DOI: [10.1051/0004-6361/202555855](https://doi.org/10.1051/0004-6361/202555855). [arXiv:2509.24593](https://arxiv.org/abs/2509.24593).
- [19] Foster, A., A. Chokshi, A. J. Anderson, et al. 2025. “Detection of Thermal Emission at Millimeter Wavelengths from Low-Earth Orbit Satellites.” *The Open Journal of Astrophysics* 8 (May): 51. DOI: [10.33232/001c.137526](https://doi.org/10.33232/001c.137526). [arXiv:2411.03374](https://arxiv.org/abs/2411.03374).
- [20] Ge, F., M. Millea, E. Camphuis, et al. 2025. “Cosmology from CMB lensing and delensed EE power spectra using 20192020 SPT-3G polarization data.” *PhRvD* 111, no. 8 (April): 083534. DOI: [10.1103 / PhysRevD.111.083534](https://doi.org/10.1103/PhysRevD.111.083534). [arXiv:2411.06000](https://arxiv.org/abs/2411.06000).
- [21] Khalife, A. R., L. Balkenhol, E. Camphuis, et al. 2025. “SPT-3G D1: Axion Early Dark Energy with CMB experiments and DESI.” *arXiv e-prints* (July): arXiv:2507.23355. DOI: [10.48550 / arXiv.2507.23355](https://doi.org/10.48550/arXiv.2507.23355). [arXiv:2507.23355](https://arxiv.org/abs/2507.23355).

- [22] Korneelje, K., L. E. Bleem, E. S. Rykoff, et al. 2025. “The SPT-Deep Cluster Catalog: Sunyaev-Zel’dovich Selected Clusters from Combined SPT-3G and SPTpol Measurements over 100 Square Degrees.” *arXiv e-prints* (March): arXiv:2503.17271. DOI: [10.48550/arXiv.2503.17271](https://doi.org/10.48550/arXiv.2503.17271). [arXiv:2503.17271](https://arxiv.org/abs/2503.17271).
- [23] Vitrier, A., K. Fichman, L. Balkenhol, et al. 2025. “Towards constraining cosmological parameters with SPT-3G observations of 25% of the sky.” *arXiv e-prints* (October): arXiv:2510.24669. DOI: [10.48550/arXiv.2510.24669](https://doi.org/10.48550/arXiv.2510.24669). [arXiv:2510.24669](https://arxiv.org/abs/2510.24669).
- [24] Zebrowski, J. A., C. L. Reichardt, A. J. Anderson, et al. 2025. “SPT-3G D1: Constraints on inflationary gravitational waves with two years of SPT-3G data.” *PhRvD* 112, no. 12 (December): 123520. DOI: [10.1103/33wm-c6pt](https://doi.org/10.1103/33wm-c6pt). [arXiv:2505.02827](https://arxiv.org/abs/2505.02827).
- [25] Ansarinejad, B., S. Raghunathan, T. M. C. Abbott, et al. 2024. “Mass calibration of DES Year-3 clusters via SPT-3G CMB cluster lensing.” *JCAP* 2024, no. 7 (July): 024. DOI: [10.1088/1475-7516/2024/07/024](https://doi.org/10.1088/1475-7516/2024/07/024). [arXiv:2404.02153](https://arxiv.org/abs/2404.02153).
- [26] Prabhu, K., S. Raghunathan, M. Millea, et al. 2024. “Testing the Λ CDM Cosmological Model with Forthcoming Measurements of the Cosmic Microwave Background with SPT-3G.” *ApJ* 973, no. 1 (September): 4. DOI: [10.3847/1538-4357/ad5ff1](https://doi.org/10.3847/1538-4357/ad5ff1). [arXiv:2403.17925](https://arxiv.org/abs/2403.17925).
- [27] Raghunathan, S., P. A. R. Ade, A. J. Anderson, et al. 2024. “First Constraints on the Epoch of Reionization Using the Non-Gaussianity of the Kinematic Sunyaev-Zel’dovich Effect from the South Pole Telescope and Herschel-SPIRE Observations.” *PhRvL* 133, no. 12 (September): 121004. DOI: [10.1103/PhysRevLett.133.121004](https://doi.org/10.1103/PhysRevLett.133.121004). [arXiv:2403.02337](https://arxiv.org/abs/2403.02337).
- [28] Tandoi, C., S. Guns, A. Foster, et al. 2024. “Flaring Stars in a Nontargeted Millimeter-wave Survey with SPT-3G.” *ApJ* 972, no. 1 (September): 6. DOI: [10.3847/1538-4357/ad58db](https://doi.org/10.3847/1538-4357/ad58db). [arXiv:2401.13525](https://arxiv.org/abs/2401.13525).
- [29] Balkenhol, L., D. Dutcher, A. Spurio Mancini, et al. 2023. “Measurement of the CMB temperature power spectrum and constraints on cosmology from the SPT-3G 2018 TT, TE, and EE dataset.” *PhRvD* 108, no. 2 (July): 023510. DOI: [10.1103/PhysRevD.108.023510](https://doi.org/10.1103/PhysRevD.108.023510). [arXiv:2212.05642](https://arxiv.org/abs/2212.05642).
- [30] Pan, Z., F. Bianchini, W. L. K. Wu, et al. 2023. “Measurement of gravitational lensing of the cosmic microwave background using SPT-3G 2018 data.” *PhRvD* 108, no. 12 (December): 122005. DOI: [10.1103/PhysRevD.108.122005](https://doi.org/10.1103/PhysRevD.108.122005). [arXiv:2308.11608](https://arxiv.org/abs/2308.11608).
- [31] Schiappucci, E., F. Bianchini, M. Aguena, et al. 2023. “Measurement of the mean central optical depth of galaxy clusters via the pairwise kinematic Sunyaev-Zel’dovich effect with SPT-3G and DES.” *PhRvD* 107, no. 4 (February): 042004. DOI: [10.1103/PhysRevD.107.042004](https://doi.org/10.1103/PhysRevD.107.042004). [arXiv:2207.11937](https://arxiv.org/abs/2207.11937).
- [2] **Chichura, P. M.**, A. Foster, C. Patel, et al. 2022. “Asteroid Measurements at Millimeter Wavelengths with the South Pole Telescope.” *ApJ* 936, no. 2 (September): 173. DOI: [10.3847/1538-4357/ac89ec](https://doi.org/10.3847/1538-4357/ac89ec). [arXiv:2202.01406](https://arxiv.org/abs/2202.01406).
- [32] Ferguson, K. R., A. J. Anderson, N. Whitehorn, et al. 2022. “Searching for axionlike time-dependent cosmic birefringence with data from SPT-3G.” *PhRvD* 106, no. 4 (August): 042011. DOI: [10.1103/PhysRevD.106.042011](https://doi.org/10.1103/PhysRevD.106.042011). [arXiv:2203.16567](https://arxiv.org/abs/2203.16567).
- [33] Ghosh, A., F. Mertens, G. Bernardi, et al. 2020. “Foreground modelling via Gaussian process regression: an application to HERA data.” *MNRAS* 495, no. 3 (January): 2813–2826. DOI: [10.1093/mnras/staa1331](https://doi.org/10.1093/mnras/staa1331). [arXiv:2004.06041](https://arxiv.org/abs/2004.06041).
- [4] Kohn, S. A., J. E. Aguirre, P. La Plante, et al. 2019. “The HERA-19 Commissioning Array: Direction-dependent Effects.” *ApJ* 882, no. 1 (September): 58. DOI: [10.3847/1538-4357/ab2f72](https://doi.org/10.3847/1538-4357/ab2f72). [arXiv:1802.04151](https://arxiv.org/abs/1802.04151).